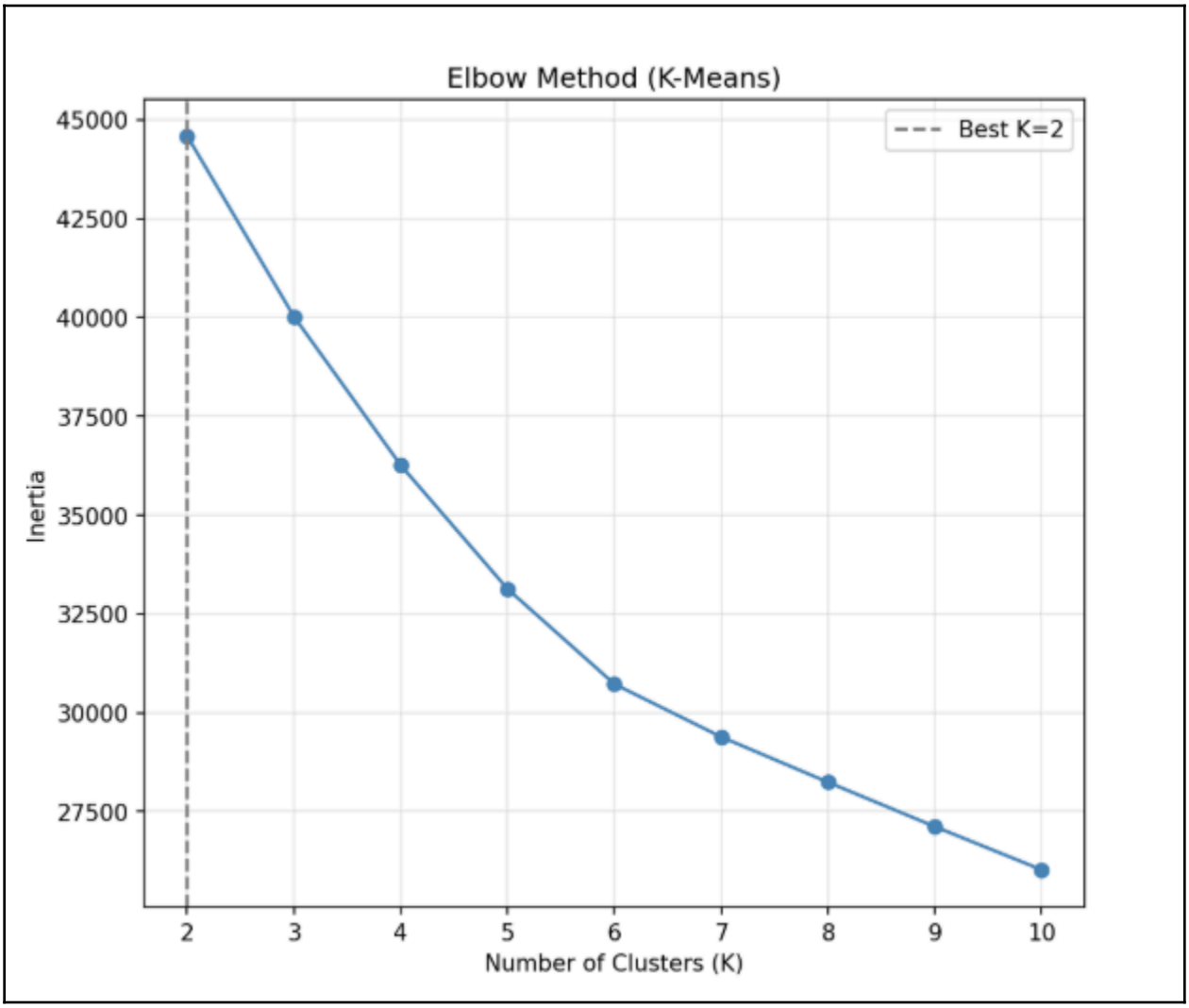
Experiment 6

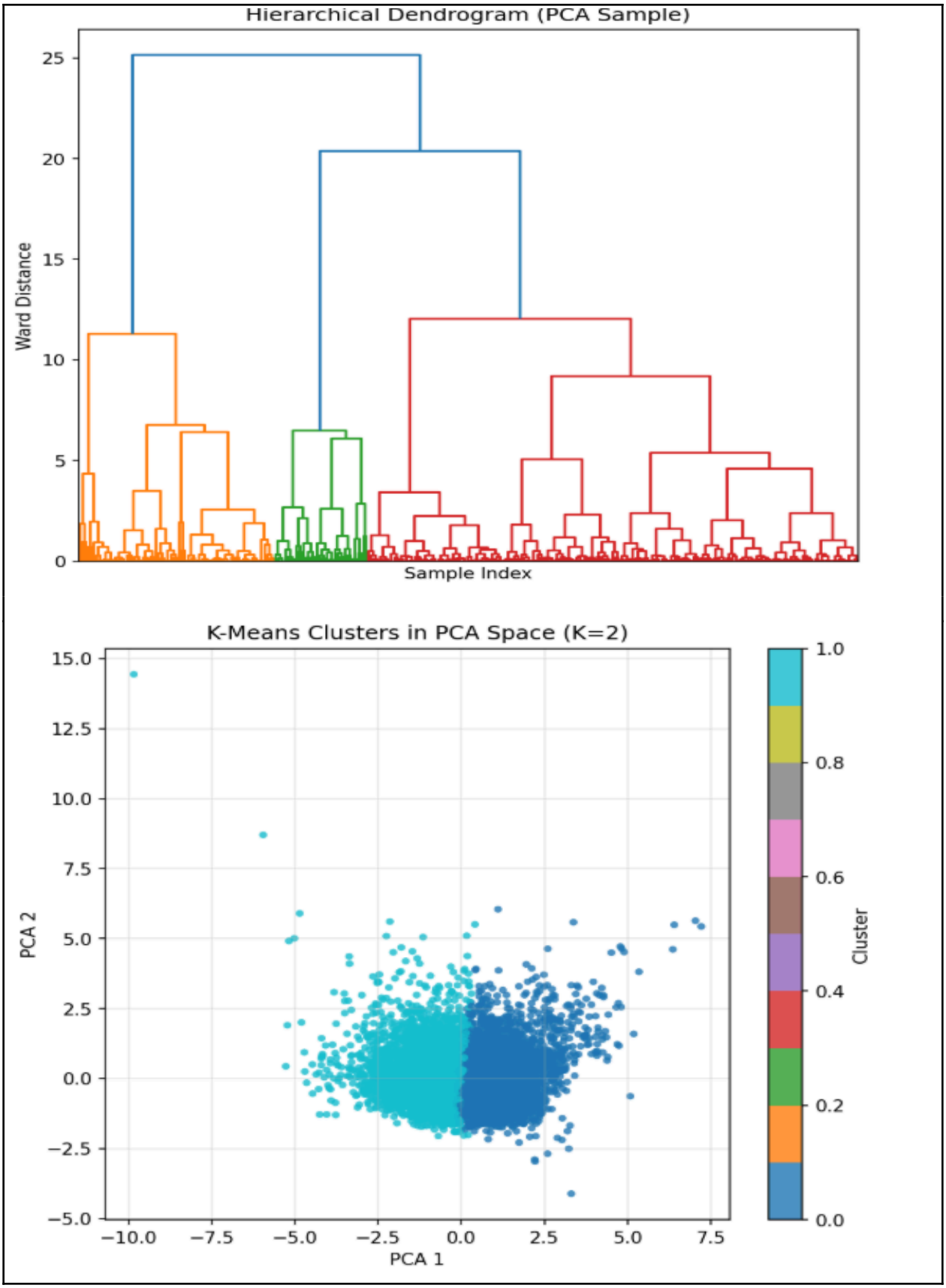
Aim : Apply K-Means and Hierarchical Clustering on sample datasets

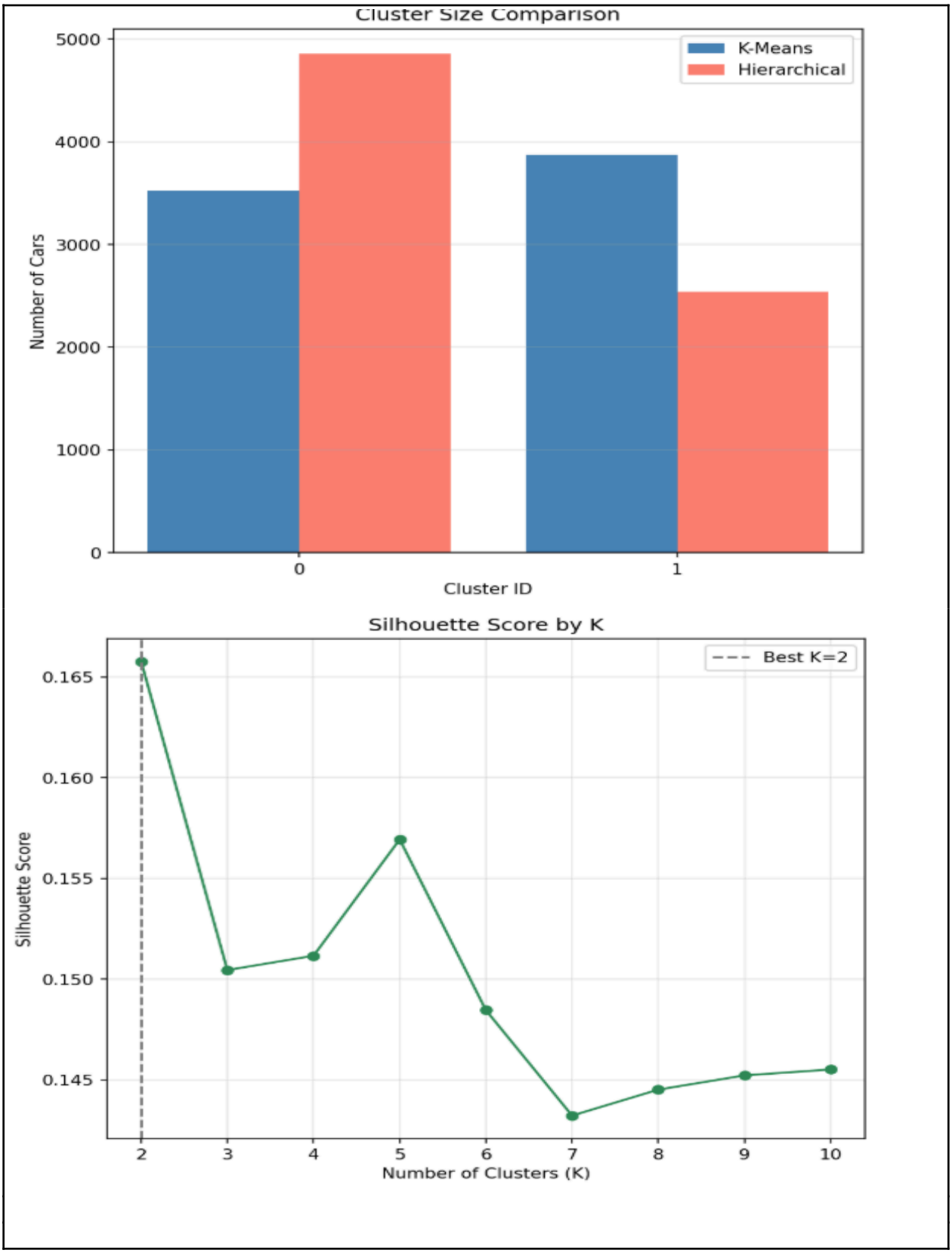
Data Source : (Used Car Price Prediction Dataset)

- Dataset Type: Structured tabular dataset
- File Format: CSV (Comma Separated Values)
- Total Records (Rows): 7,400
- Total Attributes (Columns): 29
- Domain: Used Car Sales / Automobile Marketplace Data
- [Dataset Link]

Output :




Hierarchical Dendrogram (PCA Sample)
Ward Distance
Sample Index
0
5
10
15
20
25
K-Means Clusters in PCA Space (K=2)
PCA 2
PCA 1
Cluster
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
1.0
0.8
0.6
0.4
0.2
0.0


Cluster Size Comparison
K-Means
Hierarchical
Number of Cars
5000
4000
3000
2000
1000
0
0
1
Cluster ID
Silhouette Score by K
Best K=2
Silhouette Score
0.165
0.160
0.155
0.150
0.145
2
3
4
5
6
7
8
9
10
Number of Clusters (K)

Conclusion :
K-Means and Hierarchical Clustering are powerful techniques for pattern discovery in unlabeled data.Hence,K-Means is suitable for large datasets where the number of clusters is known While Hierarchical clustering is useful when understanding the data structure and relationships is important.